Jae U YOON

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"Accelerating the future through innovative research"

UPDATED ON May 5th, 2026

EDUCATION

UNIVERSITY OF ULSAN *Mar. 2021 ~ Feb.2027 (Expected)*

B.S. IN SCHOOL OF MATERIALS SCIENCE & ENGINEERING

- **Total GPA of 3.75 / 4.5**

(Credits taken: 115/137 as of 6th semester)

YONSEI UNIVERSITY (Academic Exchange) *Mar. 2022 ~ Aug.2022*

B.S. IN SCHOOL OF MATERIALS SCIENCE & ENGINEERING

INTERNSHIP

ADVANCED ELECTRO-CHEMISTRY LAB (Prof. Sun-jeong Kim) *Sep.2022 ~ Feb.2026*

UNDERGRADUATED RESEARCHER

<Personal research project> *Mar.2025 ~ Feb.2026*

- One-step electrodeposition of Cu_2MnSnS_4 thin films on FTO glass for solar cell applications

<Government-funded research project> *Sep.2022 ~ Feb.2023*

- Study on the electro-refining process for separation and recovery of valuable metals in waste resources

AWARDS & HONORS

-AWARDS-

2025. 12 **2nd place, Capstone design competition (Prof. Eun-chae Jeon)**

Team leader & Presenter

"Reliability evaluation and lifetime modeling of foldable displays"

-HONORS-

2025. 09 Semester High Honor, Awarded to students with high achievements throughout the semester

2023. 03 Semester High Honor, Awarded to students with high achievements throughout the semester

2022. 03 Academic Exchange Scholarship, Awarded to selected students for active participation in the academic exchange program (Yonsei University)

2021. 09 Semester High Honor, Awarded to students with high achievements throughout the semester

SKILLS

English Proficiency

• 2025. 12 – OPIC IH

Certification

• 2025.01 – Six-Sigma Green Belt

Experimental Equipment

• Potentiostat / Galvanostat, SEM-EDS, XRD, Electric Furnace, AFM, UV-Vis, AAS, Four-Point Probe

Office Automation

• Microsoft software (Word, PowerPoint, Excel), Origin Pro

Development Tool

• Visual Studio code, Visual Studio, Antigravity, MATLAB

3D Modeling Tool

• Autodesk AutoCAD

ADDITIONAL INFORMATION

• Developed and visualized a computational simulation using Visual Studio code with Claude Code and Gemini Code Assist to analyze the effect of electrolyte agitation on the diffusion layer thickness within the Electrical Double Layer (EDL)

• Developed a computational simulation based on PyBaMM framework to analyze the effect of conductive additive ratio and active material composition on charge/discharge curves across cathode material variants

• Designed and machined a custom Teflon housing to standardize the exposed area of FTO glass during the electrodeposition process

• Designed and 3D-printed a multi-slot custom holder to enable the batch ultrasonic cleaning of multiple FTO glass substrates

ACTION PLAN

• Summer 2026: Applying for a research internship at UNIST laboratory to gain hands-on experience prior to graduate school enrollment